

DBI202 - PROJECT SPEAKMASTER AI

SpeakMaster AI

Chào mừng bạn quay lại!

Tên đăng nhập

Mật khẩu

Đăng nhập

Chưa có tài khoản? Đăng ký ngay tại đây

SpeakMaster

Người dùng:
truong

Tra cứu

Luyện nói

Lịch sử

Cài đặt

Chào ngày mới !

Anh - Việt Nhập từ vựng cần tra cứu...

Từ vựng trong ngày / Tìm kiếm

hello

[hə'ləʊ]

thán từ

- chào anh!, chào chị!
- này, này
- ô này! (tò ý ngạc nhiên)

danh từ

- tiếng chào
- tiếng gọi này, này !
- tiếng kêu ô này ! (tò ý ngạc nhiên)

nội động từ

- chào

world

[wɜ:ld]

danh từ

- thế giới, hoàn cầu, địa cầu

Ví dụ: to go round the world → đi vòng quanh thế giới

Ví dụ: all over the world → khắp thế giới

- vũ trụ, vạn vật

Ví dụ: since the beginning of the world → từ buổi sơ khai của vũ trụ, từ khi khai thiên lập địa

IA2001 - Mentor: NhungNTH

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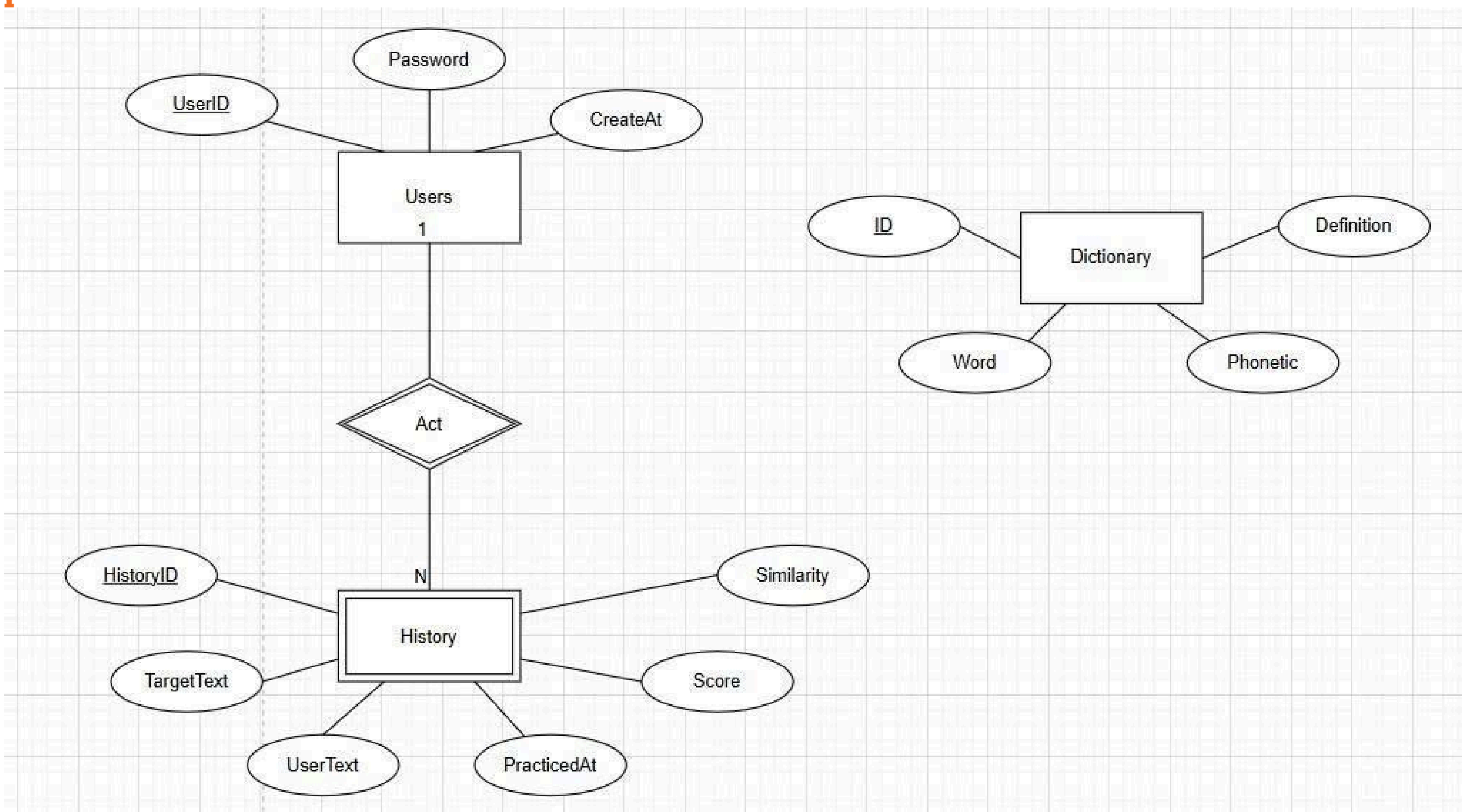
GROUP 4

NAME OF GROUP MEMBER

No.	Roll number	Fullname	Contribution percentage
1	CE200020	Nguyen Phu Anh	100%
2	CE200204	Nguyen Cao Nhut Truong	100%
3	CE200752	Tran Nhat Khoa	100%
4	CE201271	Nguyen Duc Quy	100%
5	CE201059	Le Ly Trung Hieu	100%

The system combines Speech Recognition and Natural Language Processing (NLP) to:

- Analyze speech: Use AI models (Speech-to-Text, Phonemizer) to transcribe audio and convert text into IPA representations.
- Evaluate accuracy: Automatically compare phonetic strings between user input and target sentences to provide precise pronunciation scores.
- Integrate resources: Provide a built-in English-Vietnamese dictionary for instant word lookups and vocabulary support.
- Track progress: Offer a practice history dashboard for users to review past recordings and monitor their improvement over time.



Schema Definitions:

- Users (**UserID**, Username, Password, CreatedAt)
- History (**HistoryID**, UserID, TargetText, UserText, Score, Similarity, PracticedAt)
- Dictionary (**ID**, Word, Phonetic, Definition)
- Note: Bold denotes Primary Keys (PK); Italics denote Foreign Keys (FK).

Key Relationships:

- Users 1:N History: A one-to-many relationship where one user can have multiple practice sessions.
- Referential Integrity: UserID in the History table includes an ON DELETE CASCADE constraint to prevent orphaned records.
- Standalone Reference: The Dictionary table operates independently for high-speed lookups.

Data Access Layer (Stored Procedures)

- To ensure security and modularity, all data interactions are handled via Stored Procedures:
- Authentication: sp_RegisterUser, sp_LoginUser
- Practice Management: sp_SavePracticeResult, sp_DeleteHistoryEntry
- Analytics: sp_LoadHistory, sp_GetUserStats (Aggregates average scores and progress trends).

1. AI-Powered Pronunciation Analysis

- **Speech Recognition:** Integrated OpenAI Whisper for high-accuracy speech-to-text conversion.
- **Real-time Assessment:** Instant feedback on pronunciation accuracy using the Levenshtein Distance algorithm.
- **Native Reference:** Automated speech generation via gTTS to provide standard pronunciation models.

2. Intelligent Dictionary & Search

- High-Speed Autocomplete: Optimized word suggestions using a Trie (Prefix Tree) data structure for 10,000+ words.
- Bilingual Lookup: Instant English-to-Vietnamese definitions and phonetic transcriptions retrieved from SQL Server.
- Efficient Indexing: Achieved $O(L)$ search complexity, ensuring zero-latency user interaction.

3. User Management & Analytics

- **Secure Authentication:** Personal accounts with SHA-256 encrypted passwords for data protection.
- **Performance Tracking:** Automated logging of practice history, including similarity scores and timestamps.
- **Progress Dashboard:** Visual history overview to help learners track their improvement over time

- **Languages & Libraries:** Python (Gradio for UI, pyodbc for database connectivity, hashlib for security, Whisper & gTTS for speech processing, pydub, eng_to_ipa, pandas, numpy, re, difflib).
- **Database Management:** SQL Server (Relational architecture).
- **Algorithms & Data Structures:** Trie (Prefix Tree for autocomplete), Dynamic Programming (Levenshtein Distance for pronunciation scoring).
- **Design & Modeling Tools:** Draw.io (Entity-Relationship Diagramming)

A top-down view of a desk with various items: a laptop in the upper center, a cup of coffee on the left, a pen and a highlighter in the center, a notebook on the right, three paper clips (two yellow, one teal) on the left, a pair of glasses in the center, and a large green plant leaf in the bottom left corner.

THANK YOU - Q & A